

Youssef Sghairi

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Profile

Final-year Software Engineering student at INSAT, seeking a 6-month end-of-studies internship from February 2027. Full-stack engineer shipping production features across React/Next.js, Python/Java/Go backends, and event-driven systems, owning features from design to release, with hands-on work on ALM integrations, AI-assisted workflows, and multi-tenant access control.

Experience

SpecTraceAI – Full Stack Engineer

UAE

Mar 2026 – Present

Remote

- Designed a provider-agnostic framework for REST-based ALM tools and shipped its first implementation for Polarion (bidirectional sync of requirements, test cases, and LiveDoc views); later reused to add Jama Connect with no core changes.
- Built the incoming sync pipeline: dependency-ordered snapshots (requirements, test cases and trace links, attachments), user-defined mappings for Polarion's custom work-item types, and remote deletions flagged for user approval.
- Contributed to an AI-assisted repository discovery workflow that scans monorepos directory by directory with keyword filtering to fit model context limits, proposes engineering artifacts for human review, and persists approved mappings as project configuration.
- Designed deterministic test doubles for LLM, embeddings, and vector-store integrations, improving repeatability and reducing reliance on external Azure OpenAI and Pinecone calls.
- Designed and implemented multi-tenant project support with role-based access control enforced in the service layer, enabling multiple users to securely collaborate on shared projects.

Uvey – Software Developer

Tunis, Tunisia

Jun 2025 – Mar 2026

Hybrid

- Delivered end-to-end product features using React, Quarkus, and Kafka, turning evolving client requirements and review feedback into maintainable production software through rapid iteration.
- Extracted a service from an existing microservice into a standalone service, migrating its persistence layer from MongoDB to PostgreSQL while preserving existing production data and ensuring a safe rollout.
- Built backend endpoints and business logic, resolved production issues, and contributed to architecture reviews covering service boundaries, communication patterns, migrations, and engineering trade-offs.

Selected Project

Tanilytics – Privacy-First Web Analytics Platform

TypeScript, React, Go, Spring Boot, Redpanda, ClickHouse, Redis, PostgreSQL, Kubernetes, Helm

2025 – 2026

- Co-designed (team of 3) a Go → Redpanda → ClickHouse ingestion pipeline, load-tested at 50K events/sec with synthetic traffic while tracking consumer lag; relieved the processing bottleneck with more partitions and parallel consumers.
- Implemented privacy-preserving IP handling in the processing service, before anything is stored (truncation before geolocation, then salted SHA-256 pseudonymization); duplicates are dropped at ingestion with a Redis Bloom filter.
- Built a polyglot architecture of 7 services (Go, Java/Spring Boot, TypeScript), with dashboard queries served from ClickHouse materialized views (pre-aggregated at write time) and Redis caching.
- Designed multi-tenant isolation against noisy neighbors through per-tenant rate limiting (Kong), Redis-backed hot reads, and tenant-aware ClickHouse table design.

Education

INSAT – Institut National des Sciences Appliquées et de Technologie

Engineering Degree in Software Engineering

Focus: software design, web engineering, distributed systems, databases, operating systems, algorithms, and applied machine learning.

Tunis, Tunisia

2022 – 2027

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Go

Frontend: React, Next.js, Tailwind CSS

Backend: FastAPI, Quarkus, Spring Boot

Data & Messaging: PostgreSQL, MongoDB, Redis, ClickHouse, Kafka, Redpanda, RabbitMQ

Architecture: Microservices, event-driven architecture, API design

AI & DevOps: LLM integration, RAG, Docker, Kubernetes, Helm, Github Actions

Spoken Languages: Arabic (Native), English (Full Professional), French (Full Professional), German (Elementary)

Leadership & Volunteering

INSAT ACM Student Chapter – Event Manager

Tunis, Tunisia

2024 – 2025

INSAT

- Coordinated technical events with **hundreds of participants**, managing logistics and volunteer teams.

AIESEC Eskişehir, Türkiye – Exchange Participant

Eskişehir, Türkiye

2024

- Participated in **international programs** promoting Sustainable Development Goals through education.